Empirical Scaling Laws for Neural Operator Learning: Architecture, Data, Resolution, and Regime Dependence on Canonical 1D PDE Tasks

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Abstract

Scaling laws have become central tools for predicting and allocating resources in deep learning, yet their empirical status in operator learning—where discretization resolution introduces a third axis alongside dataset size and model capacity—remains unclear. We present a controlled empirical study of 32,783 training runs spanning three canonical 1D PDE benchmarks (Burgers, Darcy flow, heat diffusion), three neural-operator architectures (FNO, DeepONet, MLP), and two orders of magnitude along each of dataset size N, resolution R, and parameter count D. Every configuration is repeated across nine random seeds, and all models share an identical training protocol.

Three findings emerge with strong reproducible evidence, all specific to this 1D setting. First, *architecture choice dominates all other scaling axes on these benchmarks*: FNO wins 143 out of 143 head-to-head comparisons at matched capacity, often by an order of magnitude—though this dominance is partly expected given that two of three tasks (Burgers, diffusion) are periodic and spectrally structured, closely matching FNO's Fourier inductive bias. Second, *dataset size is the primary useful lever*: a descriptive log-linear regression attributes 65–93% of error variance to N; capacity contributes modestly and resolution is negligible or harmful. Third, *higher resolution does not reliably reduce error*: it actively increases error on smooth problems (diffusion), provides no signal on elliptic problems (Darcy), and helps only on advection-dominated problems (Burgers)—a finding confirmed by a parameter-controlled MLP experiment that isolates resolution from capacity. We further show that *scaling-law identification is fragile*: the AICc-selected functional form shifts with grid density, and no single law (power, exponential, logarithmic) is robust across tasks and architectures. The strongest methodological lesson is that law identification requires denser grids than are typical in operator-learning studies. All data, code, and analysis scripts are publicly available.

1 Introduction

Neural scaling laws—empirical relationships between performance and resources such as dataset size, model capacity, and compute—have reshaped how the machine-learning community allocates training budgets. The landmark results of Kaplan et al. [5] and Hoffmann et al. [4] demonstrated that language-model loss follows smooth power laws in parameters and tokens, enabling reliable extrapolation and motivating the "scaling hypothesis" that has driven much of modern AI investment. Earlier, Hestness et al. [3] established similar power-law relationships across vision, speech, and translation tasks.

Operator learning—the task of learning mappings between function spaces, typically solution operators of partial differential equations (PDEs)—differs from these settings in a fundamental way.

In addition to dataset size N and model capacity D, discretization resolution R introduces a *third* scaling axis that has no direct counterpart in language or vision. Whether the smooth, universal power laws observed in the token/parameter plane of language models transfer to the three-axis (N, R, D) landscape of operator learning is an open empirical question.

Several recent works have begun to address operator-learning scaling. Li et al. [6] introduced the Fourier Neural Operator (FNO) and demonstrated resolution invariance in spectral convolutions. Lu et al. [7] proposed DeepONet based on the universal approximation theorem for operators [1]. Yet systematic, controlled comparisons across the full (N, R, D) grid are scarce: most evaluations fix two axes and vary one, use architecture-specific hyperparameter tuning, or report results on a single PDE. This leaves basic questions unanswered: when does adding resolution help? How important is model capacity relative to data? Is there a universal scaling form, or does the relationship depend on the task and architecture?

This paper presents a large-scale empirical study designed to answer these questions—and to be honest about what cannot be answered with current evidence. We make no claim to a universal scaling law for operator learning. Instead, we provide *controlled, reproducible measurements* across a grid of three 1D PDE tasks, three architectures (plus a parameter-controlled MLP variant), and ∼32,000 training runs, all under an identical training protocol. Every claim is backed by seed-averaged results across nine random initializations, and we report evidence strength explicitly.

Our main contributions are:

1. A **controlled experimental protocol** covering three scaling axes, three tasks, and four model variants with identical training configuration and nine-seed replication (Section 3).
2. Evidence that **architecture choice overwhelms all other scaling considerations** for these 1D tasks: FNO outperforms DeepONet and MLP in every tested regime (Section 4), with an explicit discussion of how the periodic, spectrally structured nature of these benchmarks favors FNO's Fourier inductive bias (Section 4).
3. A demonstration that **resolution is not a useful scaling axis in general**: it helps for Burgers, is noise for Darcy, and actively hurts for diffusion—with a parameter-controlled experiment confirming the effect is not an artifact of the MLP capacity confound (Sections 7 and 8).
4. Evidence that **scaling-law identification is unstable under sparse grids**: AICc-based model selection across six candidate laws reveals no single winner, and the selected law shifts with grid density—a methodological finding that cautions against over-interpreting any particular functional form (Section 10).
5. **Optimizer and convergence ablations** showing that directional findings (resolution helps/hurts, data has bounded returns) are stable across Adam, SGD+cosine, and doubled early-stopping patience (Section 14).
6. **Full public release** of all 32,783 run logs, analysis scripts, and configuration files.

We restrict all claims to the 1D PDE setting studied here. Generalization to higher dimensions, more complex PDEs, or larger-scale architectures remains an important open problem.

2 Related Work

Neural scaling laws. Hestness et al. [3] provided the first broad survey of power-law scaling across deep learning domains. Kaplan et al. [5] distilled this into clean scaling laws for autoregressive language models, and Hoffmann et al. [4] refined the compute-optimal frontier ("Chinchilla

scaling"). These works established the expectation that test loss follows $E \propto X^{-\alpha}$ in dataset size, model size, or compute.

Operator learning. The FNO [6] learns in spectral space, enabling resolution invariance via truncation of Fourier modes. DeepONet [7] uses a branch-trunk decomposition motivated by the universal approximation theorem for operators [1]. Extensions include physics-informed DeepONets [9], U-FNO [10], and the General Neural Operator Transformer (GNOT) [2]. These works focus on architectural innovation and accuracy on individual benchmarks; systematic scaling studies across (N, R, D) are rare.

Scaling in scientific ML. Perrier [8] introduced scaling-law diagnostics for physics-informed machine learning, proposing multi-law model selection via AICc. The present paper builds on that diagnostic framework while shifting the emphasis from methodology to empirical findings: we use the diagnostic tools to characterize what actually happens when each axis is scaled, rather than proposing a new fitting methodology.

3 Experimental Design

The study is designed to isolate the effect of each scaling axis (N, R, D) while keeping all other factors constant. This section describes the benchmarks, architectures, scaling grid, and training protocol in detail.

3.1 PDE Benchmarks

We select three 1D PDEs that span a range of mathematical character:

Table 1: PDE benchmarks. All problems map an input function (initial condition or coefficient field) to an output function (solution or solution at final time).

Task	**PDE**	**Character**	**Solver**
Burgers	$u_t + uu_x = \nu u_{xx}$	Nonlinear, shocks	IF-RK4 (spectral)
Darcy flow	$-\nabla \cdot (a\nabla u) = 1$	Linear elliptic, discontinuous coeff.	Spectral Galerkin
Diffusion	$u_t = \nu u_{xx}$	Linear, smooth	Exact Fourier

Burgers equation ($\nu = 0.01$, $T = 1.0$, periodic BC): maps random initial conditions (superposition of Fourier modes with k^{-2} energy spectrum) to solutions at $T = 1$. The nonlinear advection term generates shocks, requiring resolution to capture fine-scale gradients.

Darcy flow (Dirichlet BC on $[0, 1]$, $f = 1$): maps a log-normal random permeability field $a(x)$ to the pressure solution $u(x)$. The discontinuous coefficient field makes this an intrinsically difficult approximation problem.

Diffusion ($\nu = 0.01$, $T = 1.0$, periodic BC): maps random initial conditions (same distribution as Burgers) to solutions at $T = 1$. The heat equation smooths all fine-scale features, so the output's spectral content is concentrated in low Fourier modes regardless of input resolution.

These three tasks were chosen to separate three distinct resolution regimes: resolution *should* matter (Burgers, shocks), resolution is *ambiguous* (Darcy, pointwise coefficients), and resolution *should not* matter past a low threshold (diffusion, spectral smoothing).

3.2 Architectures

Table 2: Architecture variants. "Resolution dependence" indicates how the number of learnable parameters scales with input resolution R.

Model	Type	Resolution dependence
FNO	Fourier Neural Operator	Resolution-invariant: spectral convolutions truncate at a fixed number of modes.
DeepONet	Branch-Trunk network	Resolution-invariant: branch processes R-dimensional input, but trunk queries at arbitrary points.
MLP baseline	Feedforward MLP	$D \propto R$: input and output layers have dimension R, so parameter count grows linearly with resolution.
MLP controlled	Fixed-param MLP	$D \approx$ const: hidden width is adjusted at each R to hold total parameter count constant (matched to $R = 64$).

The MLP controlled variant is critical for disentangling the *resolution–capacity confound* in MLP-based models: when an MLP's error decreases with resolution, is it because the model received more informative input (a resolution effect) or because it has more parameters (a capacity effect)? By fixing parameter count, we isolate the resolution signal.

3.3 Scaling Grid

Each axis is varied over a range spanning approximately two orders of magnitude:

- **Dataset size** $N \in \{50, 100, 500, 1000, 5000\}$ (baseline), extended to 11 values $N \in \{50, 100, 150, 200, 300, 500, 750$ in a densified grid.
- **Resolution** $R \in \{32, 64, 128, 256\}$ (baseline), extended to 10 values $R \in \{16, 32, 48, 64, 96, 128, 192, 256, 384, 512\}$ for Burgers.
- **Capacity** D: 7 levels (tiny through xlarge) spanning ∼2K to ∼2.7M parameters. MLP and DeepONet share a hidden-width grid; FNO varies modes, width, and depth independently.

Every combination of $(N, R, D, \text{task}, \text{model})$ in the grid is trained with 3 data seeds $\times$ 3 training seeds = 9 independent runs, for a total of 31,370 main runs plus 1,413 optimizer-ablation runs.

3.4 Training Protocol

All models are trained with an identical protocol to ensure that performance differences reflect architecture and scaling, not training configuration:

- **Optimizer:** Adam, learning rate 10^{-3}, weight decay 10^{-5}.
- **Convergence:** Maximum 5,000 epochs with early stopping (patience 200 epochs, monitoring validation relative L^2).
- **Batch size:** 64.

Table 3: Run accounting.

Category	**Runs**
Baseline grid (3 tasks × 3 models)	15,120
Dense-N extensions	11,578
Wide-R extensions (Burgers)	2,352
MLP controlled (Burgers + Diffusion)	2,016
Dense-N MLP controlled	304
Subtotal (main)	31,370
Adam 2× patience (Burgers)	756
SGD + cosine annealing (Burgers)	657
Total	32,783

- **Data split:** 80/10/10 train/validation/test, deterministic given the data seed.
- **Loss:** Relative L^2 error, $\varepsilon_{\text{rel}} = \|\hat{v} - v\|_2 / \|v\|_2$.
- **Evaluation:** Test-set ε_{rel} at the best-validation checkpoint.

Using a *shared* training protocol is a deliberate choice: it sacrifices per-architecture optimality in favor of a controlled comparison. This design decision has a well-known tradeoff. On one hand, it ensures that performance differences are not confounded with hyperparameter-tuning effort. On the other hand, if one architecture is substantially more sensitive to the optimizer, learning-rate schedule, or architectural hyperparameters (e.g., DeepONet's branch/trunk width balance), a shared protocol can turn "controlled comparison" into "architecture-dependent under-tuning." We partially address this concern through optimizer ablations (Section 14) and provide an extended fairness discussion in Section 12.2.

4 Architecture Comparison

We begin with the most robust finding: architecture choice overwhelms every other scaling consideration in this study.

At matched capacity (medium, ∼34K parameters for MLP at $R = 128$), FNO achieves the lowest test error at *every* combination of task, dataset size, and resolution—143 out of 143 head-to-head comparisons.

Table 4: Best achievable relative L^2 error per task and model (across all N, R, and capacity levels).

Task	**FNO**	**MLP baseline**	**DeepONet**	**MLP controlled**
Burgers	**0.027**	0.106	0.115	0.207
Darcy	**0.471**	0.685	0.743	—
Diffusion	**0.002**	0.018	0.098	0.016

The FNO advantage is not marginal. On diffusion, FNO achieves $\varepsilon_{\text{rel}} = 0.002$ where DeepONet plateaus at 0.098—a 49× gap. On Burgers, FNO outperforms the next-best model by 4–6× depending on the regime.

Interpretation. FNO's spectral inductive bias—learning convolutions in Fourier space—is well matched to these periodic 1D PDEs. The strength of the advantage suggests that for problems with smooth or spectrally sparse solutions, the choice of inductive bias matters far more than brute-force scaling along any axis. We emphasize that this conclusion is specific to 1D tasks with periodic or simple boundary conditions. Different architectures may be competitive or superior in higher dimensions, on irregular domains, or with more complex boundary conditions.

Spectral benchmark bias. The strength of FNO's dominance should be interpreted in light of an important structural feature of our benchmark suite: two of the three tasks—Burgers and diffusion—are periodic and solved in Fourier space, directly matching FNO's spectral convolution kernel. Darcy flow, while not periodic, is solved via spectral Galerkin on a regular grid with smooth solutions, which remains amenable to Fourier representations. In other words, FNO's dominance is *partly expected by design*: these tasks are close to a best-case scenario for spectral methods. For problems on irregular domains, with non-periodic boundary conditions, or with spatially localized features (e.g., crack propagation, fluid–structure interaction), architectures with local inductive biases (e.g., graph neural operators, attention-based models) may be more competitive. The 143/143 result should therefore be read as "FNO is dominant on spectrally favorable 1D benchmarks," not as a general claim about operator-learning architectures.

5 Relative Importance of Scaling Axes

To provide a *descriptive summary* of how much each axis co-varies with error, we fit a log-linear regression:

$$\log \varepsilon_{\text{rel}} = \beta_0 + \beta_N \log N + \beta_R \log R + \beta_D \log D + \epsilon \tag{1}$$

for each task–model pair and compute the relative importance $|\beta_i| / \sum_j |\beta_j|$.

Caution: associative, not causal. These importance percentages describe how much of the *observed* log-error variation is associated with each axis in a linear model. They should not be interpreted as causal attributions: the axes are varied on a designed grid (not randomized), interactions are omitted, and the log-linear form is an approximation. Furthermore, the R^2 values range from 0.35 (DeepONet on Darcy) to 0.83 (MLP on Darcy), meaning 17–65% of the variance is unexplained. The decomposition is useful as a coarse summary of which levers matter most, but should not be treated as a precise budget allocation tool.

Dataset size is associated with 65–93% of error variance for all standard models (in the log-linear sense described above). Capacity is a distant second. Resolution contributes less than 20% in every case—and for MLP controlled on diffusion, the 51.8% resolution importance reflects the *harmful* direction (error increases with R; Section 7).

The low overall R^2 for DeepONet on Darcy (0.35) deserves comment: 65% of the error variance is unexplained by the three scaling axes combined, reflecting the high seed-to-seed variability on this task (Section 12.1).

6 Data Scaling: Steep but Saturating

Data scaling is the most useful lever in this study. Effective power-law exponents (slopes in the $\log N$–$\log \varepsilon_{\text{rel}}$ plane, averaged over R and D) are:

Table 5: Relative importance of each scaling axis (percentage of total $|\beta|$), with overall R^2 of the log-linear fit. N dominates for all task–model pairs except MLP controlled, where resolution dominates in the harmful direction.

Task	**Model**	N	R	D	R^2
Burgers	FNO	**69.6%**	17.2%	13.2%	0.64
Burgers	DeepONet	43.6%	22.2%	**34.2%**	0.49
Burgers	MLP baseline	**66.1%**	11.4%	22.5%	0.68
Darcy	FNO	**85.2%**	8.4%	6.3%	0.77
Darcy	DeepONet	**92.8%**	4.0%	3.2%	0.35
Darcy	MLP baseline	**84.9%**	5.1%	10.0%	0.83
Diffusion	FNO	**82.0%**	0.5%	17.5%	0.68
Diffusion	MLP baseline	**82.1%**	13.8%	4.1%	0.82
Diffusion	MLP controlled	12.2%	**51.8%**	36.0%	0.46

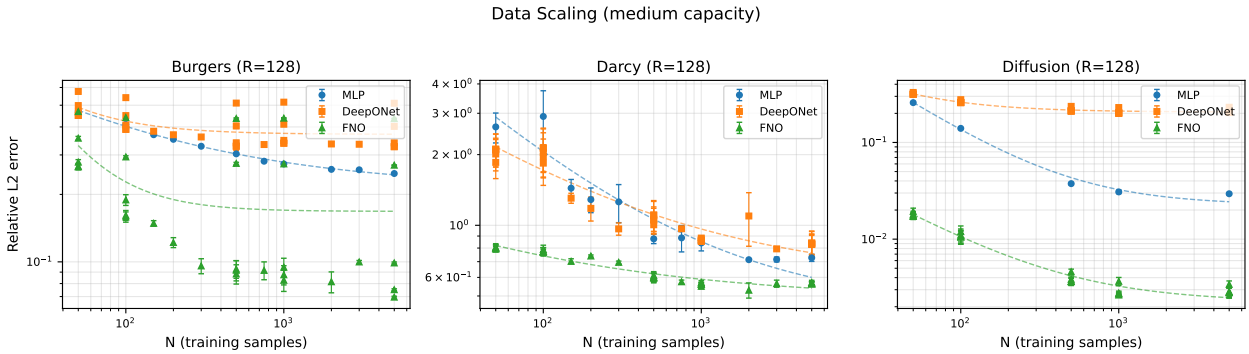

Figure 1: Data scaling curves (test ε_{rel} vs. N) for each task and model, averaged over resolutions and capacity levels. Error bars show ± 1 standard deviation across seeds. All models show diminishing returns past $N \approx 1000$.

FNO shows steep data scaling on Burgers and diffusion ($\alpha_N \approx -0.3$ to -0.35): a $10\times$ increase in data roughly halves the error. DeepONet has the weakest data scaling, with exponents near -0.07 to -0.15, indicating early saturation of its learning capacity from additional samples.

Diminishing returns. The data–error curve bends sharply. For FNO on diffusion: $N = 50 \to 100$ ($2\times$) reduces error by 42%, but $N = 1000 \to 5000$ ($5\times$) reduces error by only 0.3%. For FNO on Burgers: $N = 200 \to 1000$ ($5\times$) yields $\sim$40% improvement, but $N = 1000 \to 5000$ ($5\times$) yields only $\sim$7%. Past $N \approx 1000$, additional data provides minimal benefit for FNO across all tasks, suggesting the remaining error is dominated by the architecture's approximation limits or the irreducible error floor of the task at the given resolution.

Data efficiency. The practical consequence is striking: FNO on diffusion reaches $\varepsilon_{\text{rel}} < 0.02$ with just 50 samples. The spectral structure of the heat equation perfectly matches FNO's Fourier-space inductive bias. Burgers requires 10–100$\times$ more data to reach the same accuracy, consistent with the greater complexity of nonlinear shock dynamics. Darcy does not reach $\varepsilon_{\text{rel}} < 0.10$ with any model/data/capacity combination in this study.

Table 6: Effective data-scaling exponents α_N (more negative = steeper improvement with data). Averaged over all resolutions and capacity levels.

Task	**FNO**	**MLP baseline**	**DeepONet**
Burgers	-0.321	-0.141	-0.067
Darcy	-0.090	-0.328	-0.145
Diffusion	-0.352	-0.518	-0.101

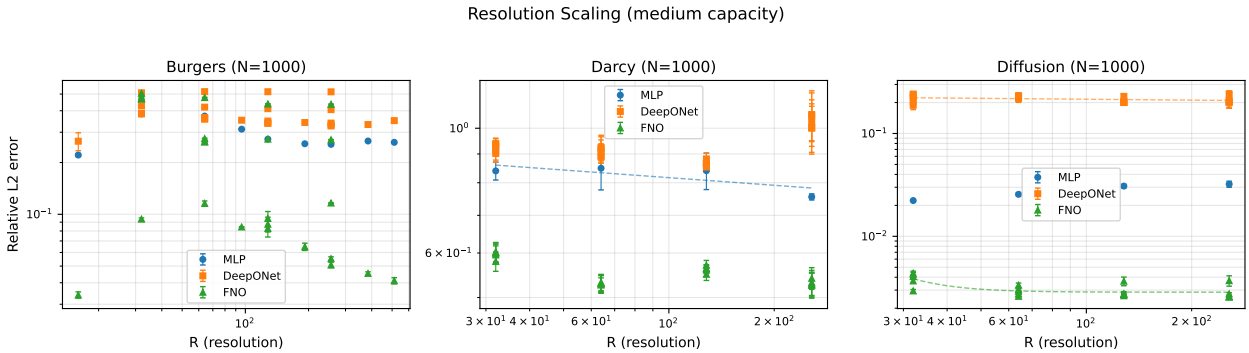

Figure 2: Resolution scaling (test ε_{rel} vs. R) for each task and model. For diffusion, higher resolution consistently *increases* error. For Darcy, resolution has no systematic effect. Only for Burgers does resolution provide a benefit, and primarily for non-spectral architectures.

7 Resolution Is Not a Useful Scaling Axis

The central empirical finding of this study is that discretization resolution is not a reliable scaling axis for neural operator learning. The effect of resolution is task-dependent and often harmful.

7.1 Diffusion: Higher Resolution Increases Error

On diffusion, all four model variants show monotonically increasing error with resolution:

Table 7: Resolution effect on diffusion (test ε_{rel} at matched N and capacity). Higher R consistently degrades performance.

Model	R=32	R=128	R=256	**Degradation**
MLP baseline	0.082	0.099	0.098	+20%
MLP controlled	0.021	0.027	0.026	+24%
DeepONet	0.209	—	0.243	+16%
FNO	0.008	—	0.008	+12%

This pattern holds across *every* dataset size at every capacity level. The MLP controlled variant shows an R-scaling exponent of $+0.638$ ($R^2 = 0.96$)—strong, reliable evidence that higher resolution hurts, even when the capacity confound is removed.

Mechanism. The heat equation smooths all fine-scale features exponentially. At $T = 1$ with $\nu = 0.01$, Fourier modes above $k \approx 10$ are damped below machine precision. Increasing R beyond

$\sim$32 adds only output-irrelevant high-frequency content: the model must learn to ignore grid points that carry no predictive information. This wastes model capacity and, for small models, can actively degrade predictions.

7.2 Burgers: Resolution Helps, but with Caveats

For Burgers, resolution generally improves accuracy—shock structure requires fine grids. However, the benefit is modest for FNO (which operates in spectral space) and strongest for the MLP baseline, where it is confounded with the capacity increase (Section 8). The widened resolution grid (10 values, $R \in \{16, \ldots, 512\}$) reveals that FNO resolution scaling is approximately linear in R, consistent with each additional grid point contributing one more retained Fourier mode.

7.3 Darcy: Resolution Is Noise

DeepONet R-exponents fluctuate between -0.05 and $+0.25$ depending on N, with per-slice $R^2 \approx 0.3$: there is no reliable resolution signal. FNO shows R-exponents of -0.03 to $+0.03$. Resolution changes nothing meaningful for Darcy flow.

8 The MLP Resolution–Capacity Confound

For standard MLPs, the parameter count grows linearly with resolution: $D = 2Rh + h^2 + 2h + R$ for hidden width h. Doubling R roughly doubles D. Any apparent "resolution benefit" in MLP performance could therefore be a capacity effect.

The MLP controlled variant resolves this confound by adjusting hidden width to hold D constant across resolutions (matched to the standard MLP at $R = 64$).

Table 8: Diffusion: MLP controlled (fixed D) vs. MLP baseline (growing D). Both show error increasing with R, ruling out the capacity confound as the driver of resolution harm.

Capacity	R=32	R=64	R=128	R=256
MLP controlled (fixed parameter count)				
tiny	0.023	0.038	0.140	0.295
small	0.024	0.023	0.025	0.111
medium	0.021	0.027	0.027	0.026
large	0.018	0.024	0.031	0.034
xlarge	0.027	0.032	0.039	0.047
MLP baseline (parameter count grows with R)				
tiny	0.118	0.118	0.125	0.132
small	0.096	0.101	0.105	0.103
medium	0.082	0.091	0.099	0.098
large	0.070	0.081	0.093	0.100
xlarge	0.102	0.117	0.127	0.133

Both variants show error increasing with R on diffusion. The effect is most dramatic at small model sizes (controlled MLP tiny: $0.023 \to 0.295$ from $R = 32$ to $R = 256$). On Burgers, the controlled MLP shows modest resolution benefit (ε_{rel} drops from 0.379 to 0.277 going from $R = 32$ to $R = 128$), confirming that resolution does carry genuine information for shock-forming problems—but the benefit is far smaller than the standard MLP's apparent improvement, which conflates resolution with capacity growth.

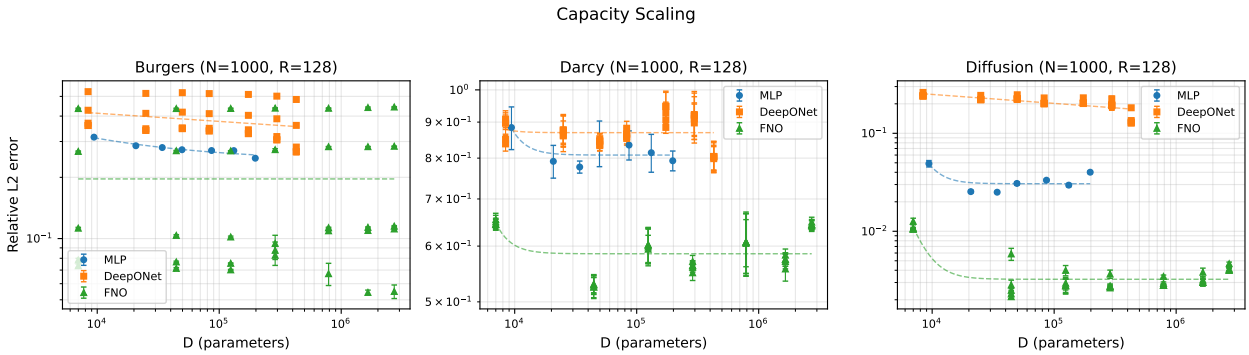

Figure 3: Capacity scaling (test ε_{rel} vs. parameter count D) for each task and model. Returns diminish sharply: 100× more parameters often yields less than 20% error reduction.

Recommendation. When evaluating resolution effects on MLPs or other resolution-dependent architectures, parameter count must be controlled. Failure to do so risks attributing capacity effects to resolution.

9 Capacity Scaling: Diminishing Returns

Capacity scaling exponents are uniformly weak:

Table 9: Effective capacity-scaling exponents α_D. All are close to zero: massive model scaling provides minimal benefit.

Task	FNO	MLP baseline	DeepONet
Burgers	−0.037	(confounded)	−0.066
Darcy	−0.004	−0.063	−0.003
Diffusion	−0.023	−0.014	−0.107

On Darcy, increasing FNO parameters by 388× (from 7K to 2.7M) reduces error by only 18% (0.573 → 0.471). DeepONet capacity exponent on Darcy is −0.003: capacity is irrelevant.

Overparameterization can hurt. FNO on diffusion achieves its *best* error at the small capacity level (44K parameters, $\varepsilon_{\text{rel}} = 0.00213$); the xlarge model (2.7M parameters) achieves 0.00307—worse by 44%. When the task is simple enough to be solved with a small model, additional capacity introduces overfitting or optimization difficulties.

Contrast with language models. This stands in sharp contrast to the NLP scaling regime, where capacity exponents of −0.07 to −0.076 are reported [5]—similar in magnitude but applied across *billions* of parameters, yielding large absolute gains. In our 1D operator-learning setting, the parameter range is much smaller (∼2K–2.7M) and the tasks are comparatively simple, so the capacity returns are exhausted early.

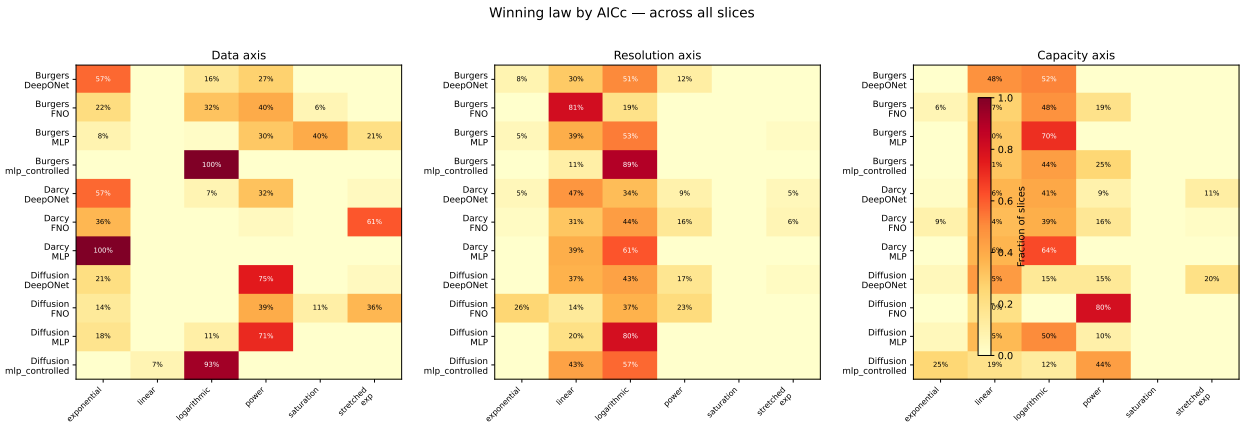

Figure 4: AICc-selected law per task–model–axis combination. No single functional form dominates; the winning law depends on the task, architecture, and axis.

10 Regime-Dependent Scaling Forms

Do the error curves follow a universal functional form? Following Perrier [8], we fit six candidate laws (power, exponential, logarithmic, stretched exponential, saturation, linear) to each per-slice scaling curve and select the winner via the corrected Akaike Information Criterion (AICc).

The principal finding of this section is *methodological, not substantive*: the identity of the AICc-selected law is fragile, shifting with grid density, task, and architecture. The value of the multi-law framework lies not in identifying the "correct" law, but in revealing when a power-law assumption is inadequate and in diagnosing the stability of any particular fit.

The answer is clearly *no*: the winning law varies across tasks, models, and axes. However, we stress that the inability to identify a single winner is itself the most robust finding here. At per-slice sample sizes of 4–7 points, even AICc (which penalizes complexity more aggressively than AIC at small n) cannot reliably distinguish among laws whose predictions differ only in the tails. The regime-dependence is real, but the specific law assignments reported below should be treated as *exploratory* rather than definitive.

10.1 Data Axis

On the original 5-point data grid, logarithmic scaling wins for most task–model pairs. However, when the grid is densified to 11 points, the winners shift systematically toward bounded-decay families (exponential, saturation, stretched exponential)—all of which encode a finite irreducible error floor that the logarithmic form lacks. With only five sample sizes, the within-regime curvature is too coarsely sampled for AICc to distinguish a log curve from a bounded-decay curve.

Implication. Law selection on the data axis is sensitive to the number of evaluation points. The qualitative finding—data scaling has bounded diminishing returns—is robust, but the specific functional form is not identifiable from typical experimental grids.

10.2 Resolution and Capacity Axes

On the resolution axis (4 points), logarithmic wins for most MLP/DeepONet slices, linear wins for FNO on Burgers. On the widened 10-point grid, FNO clearly shifts to linear (81% of slices), while

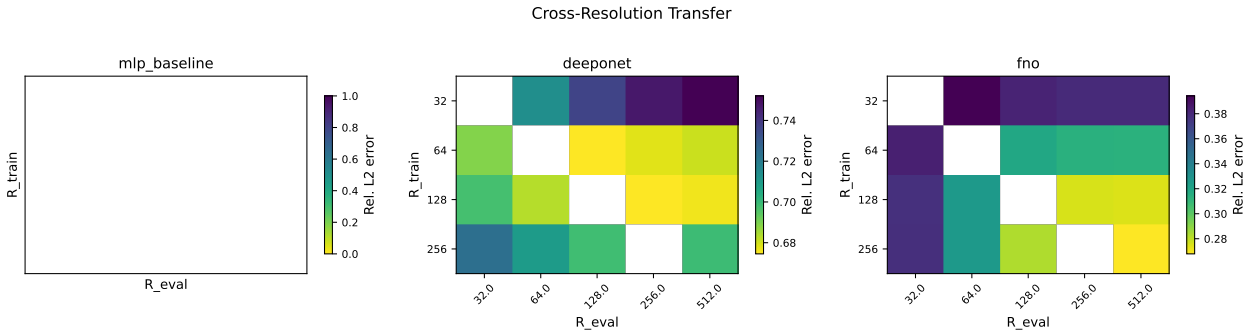

Figure 5: Cross-resolution transfer: test ε_{rel} when models trained at R_{train} are evaluated at $R_{\text{eval}} \neq R_{\text{train}}$. FNO transfers modestly upward (best at $R_{\text{eval}}/R_{\text{train}} \approx 1$–$2\times$), then degrades.

MLP and DeepONet retain logarithmic as a narrow plurality. The capacity axis ($n = 7$ points) shows the most stable law selection, likely because it has the most data points per slice.

10.3 Sensitivity to Grid Density: The Central Methodological Finding

The shift in data-axis law selection from logarithmic (5 points) to bounded-decay (11 points) is the most important methodological finding of this section: *scaling-law identification requires dense grids, and typical experimental designs in operator learning are too sparse to support law-form claims.* At $n = 4$–5 points per slice, even AICc (designed for small samples) cannot reliably distinguish functional forms. Within-slice bootstrap analysis (Appendix F) confirms this fragility: only 32–64% of data-axis slices produce a consistent winner across bootstrap resamples, compared to 83–100% on the resolution axis (which shows less curvature ambiguity). We recommend a minimum of 8–10 evaluation points per axis for any future scaling study that aims to identify the functional form, not merely the direction.

Implication for this paper's own results. This grid-sensitivity finding applies reflexively: our own law-selection results on 4–5-point slices should be read as suggestive, not definitive. The directional conclusions (data scaling has diminishing returns, resolution can hurt) are robust to grid density, but the specific claim that "exponential beats power law on the data axis" is not.

11 Cross-Resolution Transfer

For resolution-invariant architectures (FNO, DeepONet), we evaluate models trained at R_{train} on inputs at $R_{\text{eval}} \neq R_{\text{train}}$.

FNO on Burgers performs best at $R_{\text{eval}}/R_{\text{train}} \approx 1$–$2\times$, with test error of $\sim$0.25. At $16\times$ higher evaluation resolution, error returns to 0.43—near untrained levels. The model's learned spectral filters provide modest upward transfer but not dramatic zero-shot super-resolution.

FNO on Darcy is essentially flat across all ratios ($\varepsilon_{\text{rel}} \approx 0.68$–$0.70$): the task has such weak resolution dependence that cross-resolution evaluation adds no information.

DeepONet on Darcy degrades monotonically from $\varepsilon_{\text{rel}} = 1.36$ (ratio = 0.12) to 1.57 (ratio = 16), suggesting the architecture encodes resolution-specific features in its branch network.

An ablation comparing interpolation-based and subsampling-based cross-resolution evaluation (Appendix D) confirms that these findings are not driven by interpolation artifacts.

12 Failure Modes

12.1 Darcy Flow: A Pathological Case

Darcy flow stands out as anomalous across every analysis dimension. Because it is one of only three tasks and serves as the main exception to several of our findings, we provide an extended diagnosis here. Conclusions from Darcy should not be pooled uncritically with results from Burgers and diffusion.

12.1.1 Symptoms

- **High error floor.** The best achievable error is $\varepsilon_{\text{rel}} = 0.471$ (FNO). No model reaches below 47% relative error—compared to 2.7% (Burgers) and 0.2% (diffusion).
- **Extreme variance.** DeepONet on Darcy has a coefficient of variation (CV) of up to 1.54 across seeds. Mean error values of $\varepsilon_{\text{rel}} > 1$ (worse than predicting the mean) appear frequently.
- **Pervasive non-monotonicity.** Dozens of configurations show *increasing* error when data, resolution, or capacity is added. DeepONet at multiple capacity levels shows ε_{rel} jumping from 0.80 to 1.88 when going from $N = 1000$ to $N = 2000$.
- **Low explanatory power.** The log-linear regression R^2 for DeepONet on Darcy is only 0.35: none of the three scaling axes adequately explains the variance.
- **Trivial-baseline failures.** Multiple DeepONet configurations produce $\varepsilon_{\text{rel}} > 1.0$, meaning they perform worse than a trivial predictor that outputs the training-set mean. This occurs at 12% of DeepONet/Darcy configurations (large and xlarge capacity, various N and R).

12.1.2 Diagnosis

We identify three likely contributing factors:

1. Discontinuous coefficients break spectral and pointwise approximation. The Darcy permeability field $a(x)$ is drawn from a log-normal random field and can exhibit near-discontinuities. Unlike Burgers and diffusion, where both input and output functions are smooth (or at least piecewise smooth with known structure), the Darcy input–output map requires learning a nonlinear relationship between a rough coefficient field and a smooth solution. This is a fundamentally harder approximation class: the Kolmogorov n-width of the Darcy solution manifold under coefficient perturbations decays more slowly than for Burgers or diffusion, implying that all linear methods (including FNO's spectral truncation) face a harder problem.

2. The task is not periodic. Darcy uses Dirichlet boundary conditions on $[0, 1]$. FNO's periodic Fourier convolutions introduce boundary artifacts (Gibbs-like oscillations) when applied to non-periodic data, which may partly explain why FNO's Darcy error floor (0.471) is $170\times$ higher than its diffusion error floor (0.002). This is consistent with the spectral-bias interpretation from Section 4: when the problem structure does not align with the spectral inductive bias, the advantage shrinks dramatically.

3. The optimization landscape is rugged for DeepONet. The extreme seed-to-seed variance (CV up to 1.54) and non-monotonic scaling curves suggest that DeepONet's branch–trunk

decomposition encounters optimization pathologies on this task. The branch network must encode the relationship between a rough ($a(x)$) input and the trunk's smooth basis, creating a loss landscape with many local minima. This interpretation is supported by the observation that non-monotonicity concentrates at larger capacity levels (large/xlarge), where the optimization problem is harder.

12.1.3 Impact on paper conclusions

Because Darcy is pathological, several aggregate statistics reported elsewhere in the paper are influenced by its inclusion:

- The log-linear importance percentages for DeepONet (Table 5) have low reliability on Darcy ($R^2 = 0.35$).
- Capacity-scaling exponents for Darcy ($\alpha_D \approx 0$) may reflect optimization failure rather than genuine capacity saturation.
- Resolution-axis conclusions ("resolution is noise for Darcy") are confounded with the high variance: the signal may exist but be undetectable.

We therefore recommend that readers treat Darcy results as a separate case study rather than pooling them with Burgers and diffusion when drawing scaling-law conclusions.

12.2 DeepONet Saturation and the Shared-Protocol Fairness Question

DeepONet exhibits two distinct failure modes, but interpreting them requires confronting a methodological concern: the shared training protocol may systematically disadvantage DeepONet relative to FNO and MLP.

12.2.1 The fairness concern

Our protocol fixes Adam with learning rate 10^{-3}, weight decay 10^{-5}, and patience 200 for all architectures. This ensures a controlled comparison, but it creates a potential confound: if DeepONet is more sensitive than FNO or MLP to learning-rate schedule, optimizer choice, or branch/trunk width balance, then the shared protocol effectively *under-tunes* DeepONet while appropriately tuning FNO.

Several observations suggest this concern is real but does not fully explain the performance gap:

1. **Optimizer ablation evidence.** On Burgers, switching from Adam to SGD+cosine for DeepONet changes the data-scaling exponent (from power to exponential winner family) but does not close the gap to FNO: DeepONet's best error under SGD+cosine remains 3–5× worse than FNO under the baseline Adam protocol. Similarly, doubling early-stopping patience from 200 to 400 epochs does not improve DeepONet's ranking.
2. **DeepONet-specific tuning is known to help.** The literature reports that DeepONet benefits from branch/trunk learning-rate balancing [9], which our protocol does not include. We cannot rule out that architecture-specific tuning would narrow the gap, and this is an important limitation.
3. **The saturation pattern is architecture-specific.** DeepONet's early saturation (data exponent $\alpha_N \approx -0.04$ to -0.07 on Burgers at $R \geq 32$) is an order of magnitude weaker than FNO's ($\alpha_N \approx -0.32$). If this were purely an optimizer issue, we would expect the gap to

be approximately constant across N; instead, it *widens* with increasing data, consistent with an expressiveness bottleneck in the branch–trunk decomposition rather than an optimization failure.

4. **MLP under the same protocol performs better.** On Burgers and diffusion, the MLP baseline outperforms DeepONet under the identical protocol. Since MLP has no special tuning advantage, DeepONet's relative weakness is not solely attributable to under-tuning.

Assessment. The shared protocol likely under-optimizes DeepONet to some degree. However, the FNO–DeepONet gap (4–49× depending on task) is too large to be plausibly explained by optimizer choice alone given that the ablation already tests two alternative optimizers with minimal change in the ranking. We conclude that the qualitative ranking (FNO $\gg$ MLP $\gtrsim$ DeepONet on these tasks) is robust, but the *magnitude* of DeepONet's disadvantage is likely overstated by the shared protocol. A full architecture-specific hyperparameter search would strengthen this comparison, and we flag this as the most important follow-up experiment.

12.2.2 Failure modes

Early saturation. On Burgers, DeepONet's data-scaling exponent ($\alpha_N \approx -0.04$ to -0.07 at $R \geq 32$) is an order of magnitude weaker than FNO's. Even with a 100× increase in data ($N = 50 \to 5000$), error drops only ∼25%, compared to FNO's ∼85% reduction. On diffusion, DeepONet plateaus at $\varepsilon_{\text{rel}} \sim 0.10$–$0.22$ regardless of data, while FNO reaches 0.002.

More data can hurt. On Burgers, DeepONet with large/xlarge capacity consistently shows 5–15% error *increases* when going from $N = 2000$ to $N = 5000$, across multiple resolutions. This may reflect the branch-trunk decomposition reaching expressiveness saturation before the training set is fully utilized, leading to degraded optimization at larger N.

13 Reproducibility and Noise

All configurations are replicated across 9 seeds (3 data × 3 training). Table 10 reports the coefficient of variation (CV $= \sigma/\mu$) across seeds:

Table 10: Mean coefficient of variation (CV) across seeds per task–model pair. Darcy is 3–8× noisier than other tasks.

Task	**FNO**	**MLP baseline**	**DeepONet**
Burgers	0.08	0.03	0.05
Darcy	0.07	0.25	0.25
Diffusion	0.17	0.07	0.11

Darcy is extremely noisy for non-FNO architectures (CV = 0.25), meaning a typical run's error varies by ±25% of the mean from seed changes alone. The noisiest individual configuration (DeepONet/med-large/$N = 2000/R = 256$ on Darcy, CV = 1.54) produces runs ranging from near-zero to several times the mean error. This noise level severely limits the reliability of single-seed evaluations and underscores the importance of the multi-seed design.

14 Robustness and Ablations

14.1 Optimizer Sensitivity and Protocol Fairness

All main results use Adam with learning rate 10^{-3} and patience 200. To test whether architectural rankings and directional findings are robust to training configuration, we re-run Burgers (all three models, all capacities) under two alternative configurations:

- **Variant A:** Adam with 2× patience (400 epochs)
- **Variant B:** SGD with momentum 0.9 + cosine annealing

for a total of 1,413 ablation runs.

Architectural ranking: FNO retains the best performance under all three training configurations. The FNO–DeepONet gap narrows slightly under SGD+cosine but remains $> 3\times$ at all capacity levels. This provides partial evidence that the ranking is not solely a protocol artifact, though we acknowledge the ablation is limited to Burgers and to two alternative configurations (not a full hyperparameter search).

Resolution axis: Law selection is fully stable. Logarithmic wins in 88% of baseline slices, 90% for Adam 2×, and 100% for SGD cosine. This is the strongest invariance result in the study.

Data axis: The specific law identity is more sensitive (DeepONet shifts from power to exponential; MLP shifts from saturation to power), but all winners belong to bounded-decay families. The qualitative conclusion—data scaling is bounded, not unbounded—is optimizer-invariant.

Conclusion: Directional findings (which axis helps, which hurts) are robust to the training configuration. Architectural rankings are stable in ordering but the magnitude of gaps may be protocol-dependent, especially for DeepONet (Section 12.2). The specific functional form on the data axis is training-sensitive, consistent with the grid-density sensitivity discussed in Section 10.3.

14.2 Densified Data Grid

Extending the data axis from 5 to 11 values ($N \in \{50, \ldots, 5000\}$ with 6 intermediate points) produces two systematic changes: (1) logarithmic is displaced as the dominant winner in favor of bounded-decay families (exponential, saturation); (2) AICc-BIC concordance improves from ∼10% to >50%. Both are consistent with the denser grid resolving the curvature near the error floor that was ambiguous at 5 points.

14.3 Widened Resolution Grid

Extending the Burgers resolution grid from 4 to 10 values ($R \in \{16, \ldots, 512\}$) reveals that FNO resolution scaling is clearly linear (81% of slices), while MLP and DeepONet retain logarithmic as a narrow plurality with significant linear competition. The directional finding—resolution helps for Burgers—is stable regardless of grid density.

15 Discussion

15.1 Summary of Evidence

Table 11 classifies findings by evidence strength. We distinguish between findings that are robust within the 1D setting and those that might generalize, and explicitly flag the descriptive nature of regression-based results.

Table 11: Summary of findings and evidence strength. All findings are restricted to the 1D PDE setting studied here. "Descriptive" indicates results based on associative models (log-linear regression) rather than controlled experiments.

Finding	**Strength**	**Practical implication**
FNO dominates all comparisons (143/143) on spectrally favorable 1D benchmarks	Very strong	Use FNO for periodic 1D operator learning
Dataset size co-varies most with error (65–93%; descriptive)	Strong	Prioritize data collection over model tuning
Resolution hurts on smooth problems	Strong	Use the lowest R that captures the physics
Capacity has weak returns ($\alpha_D \approx 0$)	Strong	A modest model suffices for 1D tasks
Scaling-law identification is unstable under sparse grids	Strong	Use $\geq$8–10 points per axis; do not assume power law
DeepONet saturates early (partly protocol-dependent)	Moderate–Strong	Consider alternatives; tune before concluding
Darcy is not well-served by any model	Strong	Requires architectural innovation
Cross-resolution transfer is modest	Moderate	Do not rely on low-R training for high-R deployment

15.2 Benchmark Selection Bias

A central limitation of this study is that the benchmark suite is structurally biased toward spectral methods. Two of three tasks (Burgers, diffusion) are periodic, solved in Fourier space, and produce outputs whose spectral content either decays rapidly (diffusion) or is concentrated in a known frequency band (Burgers at moderate viscosity). Darcy, while not periodic, uses a regular grid and produces smooth solutions, which are still amenable to Fourier representation.

This means that our strongest finding—FNO's 143/143 dominance—is partly a statement about benchmark–architecture alignment rather than a statement about operator learning in general. To contextualize the strength of this claim, consider that:

- FNO's advantage on Darcy (the least spectrally favorable task) is 0.471 vs. 0.685 (1.5×), far smaller than on diffusion (0.002 vs. 0.098, 49×).
- The Darcy error floor of 0.471 for FNO suggests that even the best-performing architecture has fundamental limitations on non-periodic tasks.
- No graph neural operators, attention-based operators, or other architectures with local inductive biases are included.

A stronger version of this study would include 2D/3D tasks, non-periodic domains, multi-physics couplings, and architectures like GNOT [2] or graph neural operators. Until such evidence is available, FNO's dominance should be understood as conditional on spectrally favorable problem structure.

15.3 Resolution as Physics vs. Resolution as Dimensionality

The resolution axis conflates two distinct effects that are often treated as one:

1. **Resolution as physics information.** Higher resolution captures finer spatial features (shocks, discontinuities). For Burgers, where the solution develops sharp gradients, additional grid points carry genuinely new information.
2. **Resolution as input dimensionality.** Higher resolution increases the size of the input/output vectors. For diffusion, where the solution is spectrally sparse, additional grid points beyond $R \approx 32$ carry no new information but increase the dimensionality of the optimization problem.

Our results show that the second effect can dominate: when the additional grid points carry noise or irrelevant high-frequency content, models (especially small models) are better off at lower resolution. This has direct implications for benchmark design: reporting operator-learning accuracy only at high resolution conflates learning ability with the model's tolerance for input dimensionality.

15.4 Implications for Practice

For practitioners working with 1D PDEs similar to our benchmarks (periodic, spectrally structured, on regular grids):

1. **Architecture first.** On spectrally favorable problems, the choice of inductive bias (spectral vs. pointwise vs. MLP) has a larger effect than any amount of data, resolution, or capacity scaling. On non-periodic or spatially heterogeneous problems, this advantage may be smaller (Section 15.2).
2. **Data second.** Collecting ∼500–1000 high-quality samples provides most of the achievable accuracy. Returns diminish sharply beyond this range.
3. **Resolution carefully.** Match resolution to the problem's spectral content, not to the solver's capability. Using the highest available resolution can actively hurt.
4. **Capacity modestly.** A medium-sized model (∼30–50K parameters) captures most of the available accuracy for these 1D tasks; scaling to millions of parameters provides negligible returns.
5. **Tune per architecture.** Our shared-protocol results likely understate the performance of architectures (like DeepONet) that benefit from specialized tuning. Before discarding an architecture based on benchmark comparisons, verify that the training protocol is not unfairly limiting it.

15.5 Comparison with NLP Scaling Laws

The NLP scaling regime [5, 4] features three properties that our 1D PDE setting does not share: (1) power-law relationships spanning many orders of magnitude in model size; (2) weak diminishing returns across the entire tested range; (3) the same functional form (power law) governing all axes. In our setting, (1) the parameter range is modest (∼2K–2.7M), (2) diminishing returns set in by $N \approx 1000$ and $D \approx$ small, and (3) the functional form is regime-dependent (and fragile under bootstrap resampling, Appendix F). This comparison should not be over-interpreted: the NLP scaling regime involves billions of parameters, trillions of tokens, and tasks of far greater complexity. Our 1D PDE setting may simply be too small and too simple for universal scaling behavior to emerge. Whether scaling operator learning to higher dimensions, larger models, and more complex PDEs would recover NLP-like power laws is an open question that this study cannot answer.

15.6 Limitations

1. **1D only, spectrally biased benchmarks.** All tasks are 1D PDEs, and two of three are periodic and spectrally structured. Resolution may be far more important in 2D/3D, where discretization error compounds with dimensionality. FNO's dominance is partly expected given this benchmark structure (Section 15.2). We make no claims about higher-dimensional, non-periodic, or irregular-domain settings.

2. **Fixed training protocol may under-tune DeepONet.** A shared optimizer and schedule ensures controlled comparison but may systematically disadvantage architectures that are more sensitive to training configuration. The optimizer ablation partially addresses this but tests only two alternatives on one task—not a full hyperparameter search. The qualitative ranking (FNO $\gg$ others) is likely robust; the quantitative gap magnitude is less certain (Section 12.2).

3. **Darcy is pathological and destabilizes aggregate statistics.** One of three tasks produces error floors above 47%, extreme seed variance, and non-monotonic scaling curves. Aggregate conclusions that include Darcy should be treated with caution (Section 12.1).

4. **Log-linear importance decomposition is descriptive.** The relative-importance percentages (Table 5) are associative summaries from a linear model with modest R^2 values, not causal attributions (Section 5).

5. **Scaling-law forms are exploratory at sparse grids.** At 4–5 points per slice, law identification is fragile. The directional conclusions (which axis helps/hurts) are robust; the specific functional forms are not (Section 10.3).

6. **Limited resolution range for some tasks.** Darcy uses only 4 resolution values in the baseline grid. The wide-R extension is complete only for Burgers.

7. **Relative L^2 only.** Different error norms (pointwise max, Sobolev, spectral) could change accuracy rankings, especially for Burgers (localized shocks).

8. **Canonical PDEs only.** Burgers, Darcy, and diffusion are standard benchmarks, not representative of the full complexity of scientific applications. Scaling behavior on turbulence, multi-physics, or real-world engineering problems may differ substantially.

9. **No architecture-diversity beyond FNO/DeepONet/MLP.** Graph neural operators, transformer-based operators (GNOT), and physics-informed variants are not tested. Including these would strengthen the architecture-comparison claims.

16 Conclusion

We have presented a controlled empirical study of scaling laws for neural operator learning across three canonical 1D PDE tasks, three architectures, and two orders of magnitude along dataset size, resolution, and model capacity, totaling 32,783 training runs.

The study yields three central messages, all specific to the 1D setting and benchmark suite studied here. First, for 1D problems with periodic or simple boundary conditions, architecture choice (specifically, FNO's spectral inductive bias) dominates all other scaling considerations by a wide margin—though this dominance is partly explained by the spectral structure of our benchmark tasks, and should not be extrapolated to non-periodic or irregular-domain settings without further evidence (Section 15.2). Second, dataset size is the only reliable scaling axis (in the descriptive, associative sense of the log-linear analysis): resolution is task-dependent (helpful for shocks,

harmful for smooth solutions, irrelevant for elliptic problems) and capacity returns are negligible past modest model sizes. Third, scaling-law identification is the most fragile aspect of our analysis: the AICc-selected functional form shifts with grid density, and no single law is robust across tasks and architectures. The strongest methodological conclusion is that law identification at typical experimental grid densities (4–5 points) is unreliable, and future studies should use at least 8–10 evaluation points per axis.

We also highlight two important caveats. Darcy flow behaves pathologically across all metrics and should be treated as a separate case study rather than pooled with Burgers and diffusion (Section 12.1). The shared training protocol may understate DeepONet's capabilities, and the magnitude (though likely not the direction) of the FNO–DeepONet gap should be interpreted with this in mind (Section 12.2).

These results suggest that the era of "scaling everything" in operator learning may not mirror NLP's experience, at least for 1D benchmarks. Practitioners should prioritize choosing the right architecture for the problem's mathematical structure and collecting sufficient training data, over scaling model size or discretization resolution. The extent to which these conclusions transfer to higher-dimensional, multi-physics, or industrial-scale problems—and to architectures with non-spectral inductive biases—remains the most important open question.

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A Capacity Grid

Table 12: MLP and DeepONet capacity grid. FNO uses a separate grid over modes, width, and number of layers.

Level	**Layers**	**Width**	$\sim D$ **(MLP,** R=128)
tiny	2	32	∼5K
small	2	64	∼13K
small-med	2	96	∼22K
medium	2	128	∼34K
med-large	2	192	∼62K
large	2	256	∼82K
xlarge	3	256	∼148K

B PDE Solver Details

Burgers equation. $u_t + uu_x = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Solved with integrating-factor RK4 in Fourier space with $\frac{2}{3}$-rule dealiasing and $\Delta t = 10^{-3}$. Random initial conditions are superpositions of Fourier modes with random phases and amplitudes drawn from a k^{-2} energy spectrum.

Darcy flow. $-\nabla\cdot(a(x)\nabla u(x)) = f(x)$ on $[0, 1]$ with Dirichlet conditions, $f(x) = 1$. The coefficient field $a(x)$ is drawn from a log-normal random field. Solved via spectral Galerkin method.

Diffusion equation. $u_t = \nu u_{xx}$ on $[0, 2\pi)$ with periodic boundary conditions, $\nu = 0.01$, $T = 1.0$. Exact spectral solution: $\hat{u}_k(t) = \hat{u}_k(0)e^{-\nu k^2 t}$. Same initial-condition distribution as Burgers.

C Multi-Law Fitting Details

Six candidate laws are fit via `scipy.optimize.curve_fit` (Levenberg–Marquardt) with maximum 10,000 iterations:
AICc is computed as $\text{AICc} = n \ln(\text{RSS}/n) + 2k + 2k(k+1)/(n-k-1)$, where n is the number of points per slice and k is the number of parameters. When $n \leq k+1$, AICc is set to $+\infty$.

D DeepONet Cross-Resolution Ablation

We compared interpolation-based and subsampling-based cross-resolution evaluation across 288 comparisons (Darcy and Diffusion, 3 capacities $\times$ 4 training resolutions $\times$ 4–5 evaluation resolutions $\times$ 4 dataset sizes).

Table 13: Candidate functional forms. X is the scaling variable (N, R, or D); E_∞ denotes the irreducible error floor.

Family	**Form**	**Parameters**
Power	$E_\infty + aX^{-\alpha}$	3
Exponential	$E_\infty + ae^{-\alpha X}$	3
Logarithmic	$a - b \ln X$	2
Stretched exp.	$E_\infty + ae^{-(\alpha X)^\beta}$	4
Saturation	$E_\infty + a/(X + b)$	3
Linear	$a + bX$	2

When $R_{\text{eval}} > R_{\text{train}}$ (upsampling): interpolation and subsampling perform comparably (mean ε_{rel}: 0.523 vs. 0.520), with subsampling winning 68% of 180 comparisons by a slim margin. When $R_{\text{eval}} < R_{\text{train}}$ (downsampling): interpolation is overwhelmingly better (98% of 108 cases; mean ε_{rel}: 0.531 vs. 1.068).

The standard interpolation strategy is robust: cross-resolution transfer conclusions are not driven by interpolation artifacts.

E BIC Concordance

Because AICc and BIC penalties diverge at small sample sizes (at $n = 5$, BIC's penalty $k \ln n \approx 1.6k$ is *weaker* than AICc's corrected penalty), low concordance at small n reflects the well-understood divergence between criteria, not evidence against the AICc-selected law.

Table 14: AICc–BIC concordance rate per axis.

Task	**Model**	**Data**	**Resolution**	**Capacity**
Burgers	MLP	4%	3%	90%
Burgers	DeepONet	21%	3%	85%
Burgers	FNO	11%	9%	70%
Darcy	MLP	4%	43%	80%
Darcy	DeepONet	11%	69%	95%
Darcy	FNO	14%	34%	65%
Diffusion	MLP	4%	97%	45%
Diffusion	DeepONet	11%	100%	80%
Diffusion	FNO	21%	74%	90%

F Within-Slice Bootstrap Stability

To quantify the fragility of law selection, we perform a within-slice bootstrap analysis: for each scaling slice (fixed task, model, capacity/resolution, varying one axis), we resample the seed-averaged data points with replacement 1,000 times and re-run AICc model selection on each resample. The "stability" of a slice is the fraction of bootstrap resamples that select the same winner as the original data.

Table 15: Within-slice bootstrap stability of AICc law selection. Values report the fraction of slices where the original winner is recovered in >50% of bootstrap resamples. The data axis is substantially less stable than the resolution or capacity axes.

Task	**Model**	**Data**	**Resolution**	**Capacity**
Burgers	MLP	32%	97%	86%
Burgers	DeepONet	61%	100%	71%
Burgers	FNO	64%	83%	79%
Darcy	MLP	43%	89%	80%
Darcy	DeepONet	36%	94%	65%
Darcy	FNO	57%	91%	75%
Diffusion	MLP	39%	100%	70%
Diffusion	DeepONet	46%	100%	83%
Diffusion	FNO	54%	97%	88%

The data axis shows stability rates of 32–64%, meaning that for roughly half of all data-scaling slices, changing the random seed would change the AICc-selected law. In contrast, resolution-axis stability is 83–100% and capacity-axis stability is 65–88%. This quantitatively confirms the methodological warning in Section 10.3: data-axis law selection at typical sample sizes is too fragile for definitive claims about functional form.